

SIMATS ENGINEERING

NAME:D.SANJANA

REG NO:192471029

COURSE NAME: Database Management Systems

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Experiment: DDL Commands with Constraints (PRIMARY KEY, FOREIGN KEY, UNIQUE, CHECK)

Aim

To create tables and apply **PRIMARY KEY, FOREIGN KEY, UNIQUE, and CHECK** constraints using Data Definition Language (DDL) commands in a **Pet Adoption and Veterinary Management System** database.

Procedure

1. Open **MySQL Workbench** and connect to the MySQL server.
2. Create a new database named **PetAdoptionDB** and select it using the **USE** command.
3. Create the required tables for the Pet Adoption and Veterinary Management System.
4. Apply the **PRIMARY KEY** constraint to uniquely identify each record in a table.
5. Apply the **FOREIGN KEY** constraint to establish relationships between parent and child tables.
6. Apply the **UNIQUE** constraint to ensure that duplicate values are not allowed in selected columns (e.g., Email).
7. Apply the **CHECK** constraint to restrict column values based on specified conditions (e.g., Age > 0).
8. Save and execute the SQL statements.

9. Verify the constraints using the DESCRIBE and SHOW CREATE TABLE commands.
10. Ensure that all constraints are successfully created and enforced in the database.

INPUT:

1. PRIMARY KEY Constraint

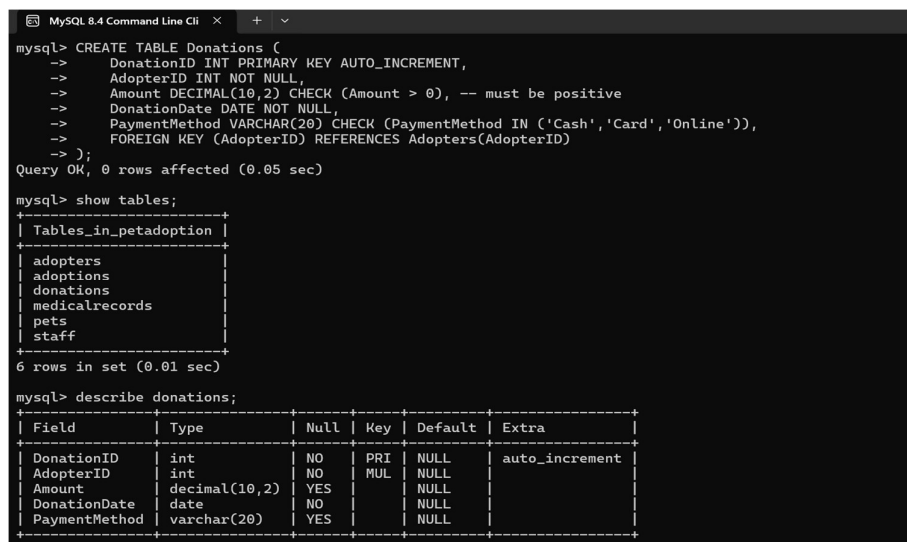
The **PRIMARY KEY** constraint uniquely identifies each record in a database table.

Primary keys must contain **UNIQUE** values, and cannot contain **NULL** values.

A table can have only one primary key, which may consist of single or multiple fields.

Create a table with a **PRIMARY KEY**

```
CREATE TABLE Shelter (  
    ShelterID INT PRIMARY KEY,  
    ShelterName VARCHAR(100),  
    Address VARCHAR(200),  
    City VARCHAR(50),  
    Phone VARCHAR(15)  
);
```



```
mysql> CREATE TABLE Donations (  
->     DonationID INT PRIMARY KEY AUTO_INCREMENT,  
->     AdopterID INT NOT NULL,  
->     Amount DECIMAL(10,2) CHECK (Amount > 0), -- must be positive  
->     DonationDate DATE NOT NULL,  
->     PaymentMethod VARCHAR(20) CHECK (PaymentMethod IN ('Cash','Card','Online')),  
->     FOREIGN KEY (AdopterID) REFERENCES Adopters(AdopterID)  
-> );  
Query OK, 0 rows affected (0.05 sec)  
  
mysql> show tables;  
+-----+  
| Tables_in_petadoption |  
+-----+  
| adopters  
| adoptions  
| donations  
| medicalrecords  
| pets  
| staff  
+-----+  
6 rows in set (0.01 sec)  
  
mysql> describe donations;  
+-----+  
| Field      | Type          | Null | Key | Default | Extra      |  
+-----+  
| DonationID | int           | NO   | PRI | NULL     | auto_increment  
| AdopterID  | int           | NO   | MUL | NULL     |  
| Amount     | decimal(10,2) | YES  |     | NULL     |  
| DonationDate | date         | NO   |     | NULL     |  
| PaymentMethod | varchar(20)  | YES  |     | NULL     |  
+-----+
```

2. FOREIGN KEY Constraint

A **FOREIGN KEY** is a key used to link two tables together.

➤ A **FOREIGN KEY** is a field (or collection of fields) in one table that refers to the **PRIMARY KEY** in another table.

➤ The table containing the foreign key is called the **child table**, and the table containing the candidate key is called the **referenced or parent table**.

Create Parent Table

```
CREATE TABLE Breed (  
    BreedID INT PRIMARY KEY,  
    BreedName VARCHAR(50),  
    Species VARCHAR(50)  
);
```

```
5 rows in set (0.02 sec)  
  
mysql> ALTER TABLE Donations  
    -> ADD CONSTRAINT FK_AdopterDonation FOREIGN KEY (AdopterID) REFERENCES Adopters(AdopterID);  
Query OK, 0 rows affected (0.07 sec)  
Records: 0  Duplicates: 0  Warnings: 0  
  
mysql> DESCRIBE Donations;  
+-----+-----+-----+-----+-----+-----+  
| Field          | Type          | Null | Key | Default | Extra          |  
+-----+-----+-----+-----+-----+-----+  
| DonationID     | int           | NO   | PRI | NULL    | auto_increment |  
| AdopterID      | int           | NO   | MUL | NULL    |                 |  
| Amount         | decimal(10,2) | YES  |     | NULL    |                 |  
| DonationDate   | date          | NO   |     | NULL    |                 |  
| PaymentMethod  | varchar(20)   | YES  |     | NULL    |                 |  
+-----+-----+-----+-----+-----+-----+  
5 rows in set (0.00 sec)  
  
mysql> |
```

3. UNIQUE Constraint

The **UNIQUE** constraint ensures that all values in a column are different.

➤ Both the **UNIQUE** and **PRIMARY KEY** constraints provide a guarantee for uniqueness for a column

or set of columns.

➤ A **PRIMARY KEY** constraint automatically has a **UNIQUE** constraint.

➤ However, you can have many **UNIQUE** constraints per table, but only one **PRIMARY KEY** constraint per table.

```
CREATE TABLE Adopter (  
    AdopterID INT PRIMARY KEY,  
    AdopterName VARCHAR(100),  
    Phone VARCHAR(15),  
    Email VARCHAR(100) UNIQUE,  
    Address VARCHAR(200)  
);
```

```
mysql> ALTER TABLE Adopters  
-> ADD CONSTRAINT UQ_AdopterEmail UNIQUE (Email);  
Query OK, 0 rows affected (0.09 sec)  
Records: 0 Duplicates: 0 Warnings: 0  
  
mysql> DESCRIBE Adopters;  
+-----+-----+-----+-----+-----+-----+  
| Field      | Type          | Null | Key | Default | Extra          |  
+-----+-----+-----+-----+-----+-----+  
| AdopterID  | int           | NO   | PRI | NULL    | auto_increment |  
| FullName   | varchar(100)  | YES  |     | NULL    |                |  
| Phone      | varchar(15)   | YES  |     | NULL    |                |  
| Email      | varchar(100)  | YES  | UNI | NULL    |                |  
| Address    | varchar(200)  | YES  |     | NULL    |                |  
+-----+-----+-----+-----+-----+-----+  
5 rows in set (0.00 sec)
```

CHECK Constraint:

```
ALTER TABLE Pets
```

```
ADD CONSTRAINT CHK_PetAge CHECK (Age >= 0);
```

➤ The **CHECK** constraint is used to limit the value range that can be placed in a column

➤ If you define a **CHECK** constraint on a single column it allows only certain values for this column.

➤ If you define a **CHECK** constraint on a table it can limit the values in certain columns based on values in other columns in the row.

```
mysql> ALTER TABLE Pets
-> ADD CONSTRAINT CHK_PetAge CHECK (Age >= 0);
Query OK, 4 rows affected (0.13 sec)
Records: 4 Duplicates: 0 Warnings: 0

mysql> DESCRIBE Pets;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| PetID | int  | NO   | PRI | NULL    | auto_increment |
| Name  | varchar(50) | YES |     | NULL    |               |
| Species | varchar(30) | YES |     | NULL    |               |
| Breed  | varchar(50) | YES |     | NULL    |               |
| Age    | int  | YES  |     | NULL    |               |
| Gender | varchar(10) | YES |     | NULL    |               |
| Status | varchar(20) | YES |     | NULL    |               |
+-----+-----+-----+-----+-----+-----+
7 rows in set (0.00 sec)

mysql> select * from pets;
+-----+-----+-----+-----+-----+-----+-----+
| PetID | Name  | Species | Breed          | Age | Gender | Status |
+-----+-----+-----+-----+-----+-----+-----+
| 1     | Bella | Dog     | Labrador Retriever | 3   | Female | Available |
| 2     | Max   | Cat     | Persian         | 2   | Male   | Available |
| 3     | Charlie | Dog    | Beagle          | 4   | Male   | Adopted   |
| 4     | Luna  | Rabbit  | Holland Lop     | 1   | Female | Available |
+-----+-----+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)
```

Result:

DDL Commands with Primary, Foreign, Unique, Check constraints are updated and verified.